A compatibility obstruction to iterating geometric HNN extensions

Human Compression project

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Piedade–Tranchida, arXiv:2505.24662v1, Section 5.4, proposes graphs of coset incidence systems and asks whether the amalgamation and HNN operations give a fundamental geometry independent of the spanning tree. The following example isolates an earlier obstruction: locally admissible edge maps need not remain admissible after the first HNN operation. It concerns the stated iterative procedure, not the impossibility of defining a different fundamental geometry.

Proposition 1. *There is a graph with one vertex and two loops, all of whose vertex and edge geometries are the same finite regular hypertope and all of whose edge identifications are admissible, for which neither ordering of the two prescribed geometric HNN operations is admissible throughout.*

Proof. Let $A = C_2^3$ with basis e_1, e_2, e_3, and give it the standard coset geometry with maximal parabolics

$$A_i = \langle e_j \,:\, j \neq i \rangle \quad (i = 1, 2, 3).$$

This is the join of three two-point geometries, hence is thin, residually connected and flag-transitive, with trivial Borel. The graph has this geometry at its sole vertex and at both loops. On each loop take one boundary map to be the identity. Take the other to be respectively the coordinate transpositions $\sigma = (12)$ and $\tau = (23)$. These are actual automorphisms permuting every indexed parabolic, so each edge map is admissible.

Perform the σ operation first and call its stable letter t. The resulting group is $G = A \rtimes_\sigma \langle t \rangle$, with t of infinite order. The type classes and maximal parabolics prescribed by the source construction are

$$\{1,2\}, \quad \{3\}, \quad *, \qquad G_{\{1,2\}} = \langle e_3, t \rangle, \quad G_{\{3\}} = \langle e_1, e_2, t \rangle, \quad G_* = A.$$

The standard parabolics contained in A are exactly

$$A, \qquad \langle e_3 \rangle, \qquad \langle e_1, e_2 \rangle, \qquad 1.$$

Indeed every intersection not involving G_* contains the infinite-order element t, and the intersections involving it are precisely the four groups displayed. The second edge still identifies the parabolic A with itself by τ, but

$$\tau(\langle e_3 \rangle) = \langle e_2 \rangle$$

is not on this list. Thus τ is not admissible for the new geometry.

If the τ operation is performed first, its two old type classes are $\{1\}$ and $\{2, 3\}$. The proper nontrivial parabolics in its A residue are $\langle e_1 \rangle$ and $\langle e_2, e_3 \rangle$. The remaining map σ sends $\langle e_1 \rangle$ to $\langle e_2 \rangle$, again absent. Hence the reverse ordering also fails. Both transpositions are their own inverses, so reversing loop orientation does not repair either order. □

The abstract fundamental group still exists: it has presentation

$$\langle A, t, u \mid t^{-1} a t = \sigma(a),\ u^{-1} a u = \tau(a)\ (a \in A) \rangle \cong A \rtimes F(t, u).$$

The obstruction is in the parabolic geometry retained after one operation. A simultaneous construction, a common refinement of the retained incidence data, or a global choice of type classes might avoid it; none is supplied by this example. The source question must therefore distinguish an unrestricted iterative recipe from a construction with additional coherent type data. This private draft has not had an independent review and carries no resolved-conjecture credit.